

Luckie Musngi

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EDUCATION

University of Arizona (Expected May 2026)

Tucson, AZ

- **Major:** Computer Science BS, **Minor:** Mathematics
- **GPA:** 3.4
- **Coursework:** Web Programming, Operating Systems, Algorithms, Database Design, Computer Networks

TECHNICAL SKILLS

Programming: C#, Python, Java, JavaScript, HTML/CSS, C, MIPS, C++

Frameworks / Tools: React.js, Unity, NumPy, Pandas, Matplotlib, Scikit-learn, Git/GitHub, REST APIs

Other skills: Agile Scrum, Embedded Systems, Hardware–Software Integration, Rapid Prototyping, Real-Time Control, Sensors, Servo/DC Motors, CAD (Fusion 360, OpenSCAD), Mechanical Design

EXPERIENCE

University of Arizona

Tucson, AZ

Undergraduate Researcher, Bio Robotics & Scientific Instrumentation

01/2026–Present

- Designing and building robotic systems to automate practical biology and chemistry experiments.
- Applying experimental design principles to evaluate system performance, reliability, and repeatability.
- Integrating motors, actuators, and sensors to implement automated experimental workflows.
- Documenting experiments and author technical reports summarizing design decisions and results.

Vail Academy and High School

Tucson, AZ

Site Tech Coordinator Shadow

09/2021–11/2021

- Gained hands-on experience troubleshooting hardware, software, and network issues; configured technical equipment for events and solved day-to-day problems.
- Learned about IT support workflows, including ticketing systems, troubleshooting protocols, and general problem-solving while collaborating with staff and faculty.

Lin's Grand Buffet

Tucson, AZ

Expeditor

07/2022–Present

- Collaborating with the team to maintain stocked and presentable areas while coordinating specialists for efficient buffet operations.
- Managing and resolving customer requests, communicating with teams in a fast-paced environment.

PROJECTS

High-Speed Arduino Sorting Machine | https://github.com/LuckieMusngi/Arduino-Skittle-Sorter_Quannel-Method

- **The World's Fastest** (~4.44 skittles/sec)
- Designed and built solo, a fully custom, high-throughput color-based sorting system using Arduino.
- Engineered a unique four-way funnel system and circular trapdoor mechanism enabling parallel sorting.
- Designed, modeled, and 3D-printed all components in OpenSCAD.
- Integrated multiple RGB sensors, servo motors, and a geared DC motor for real-time classification and routing. Optimized timing, throughput, and mechanical–software coordination.

Arduino, Embedded C/C++, OpenSCAD, Sensors, Servo/DC Motors, Rapid Prototyping, Real-Time Control, Parallel Processing

Mock Ski Resort SQL Database System | <https://github.com/LuckieMusngi/Ski-Resort-Database>

- Co-developed (2-person) a full Oracle SQL database system with Java JDBC interface.
- Planned and designed relational schema, constraints, and required queries given an application domain.
- Built a command-line interface to query and modify the database (add/update/delete operations), and implemented all table creation and data population scripts, handling all debugging and validation.

Oracle SQL, Java, JDBC, Relational Schema Design, E-R Diagrams, DDL/DML, CLI, Database Constraints

Crime in Tucson Analysis | <https://github.com/LuckieMusngi/Crime-in-Tucson-Analysis>

- Analyzed official Tucson crime data to identify patterns and predict crime levels.
- Preprocessed and cleaned data using NumPy and Pandas.
- Applied machine learning linear models and classification techniques; evaluated models using cross-validation and confusion matrices, and visualized trends and model outputs with Matplotlib.

Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, Scikit-learn, Linear Algebra, Probability, Machine Learning, Data Preprocessing, Feature Engineering, Model Evaluation

CLUBS AND ORGANIZATIONS

Software Engineering Wildcats

Member

08/2022–Present

UA Climbing Comp Team

Member

10/2024–Present